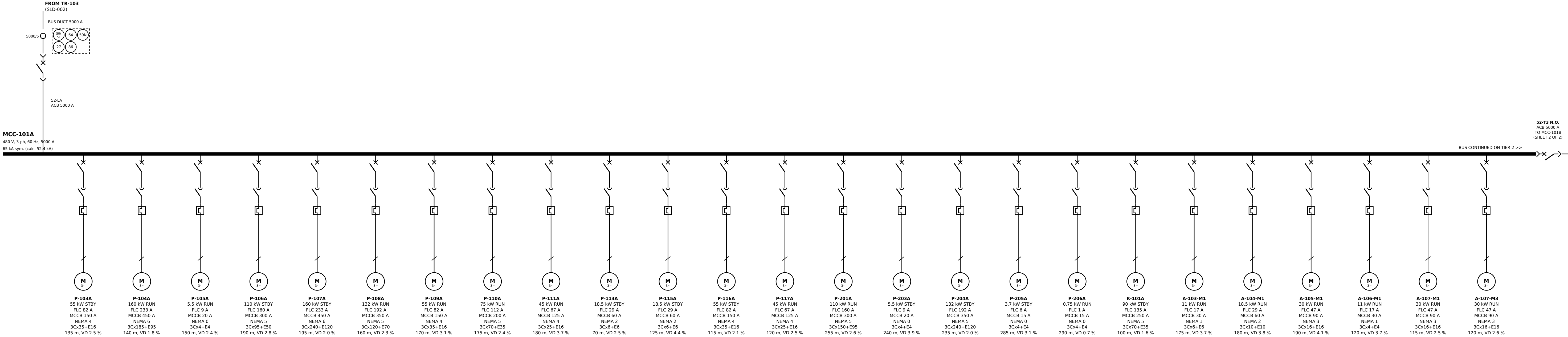
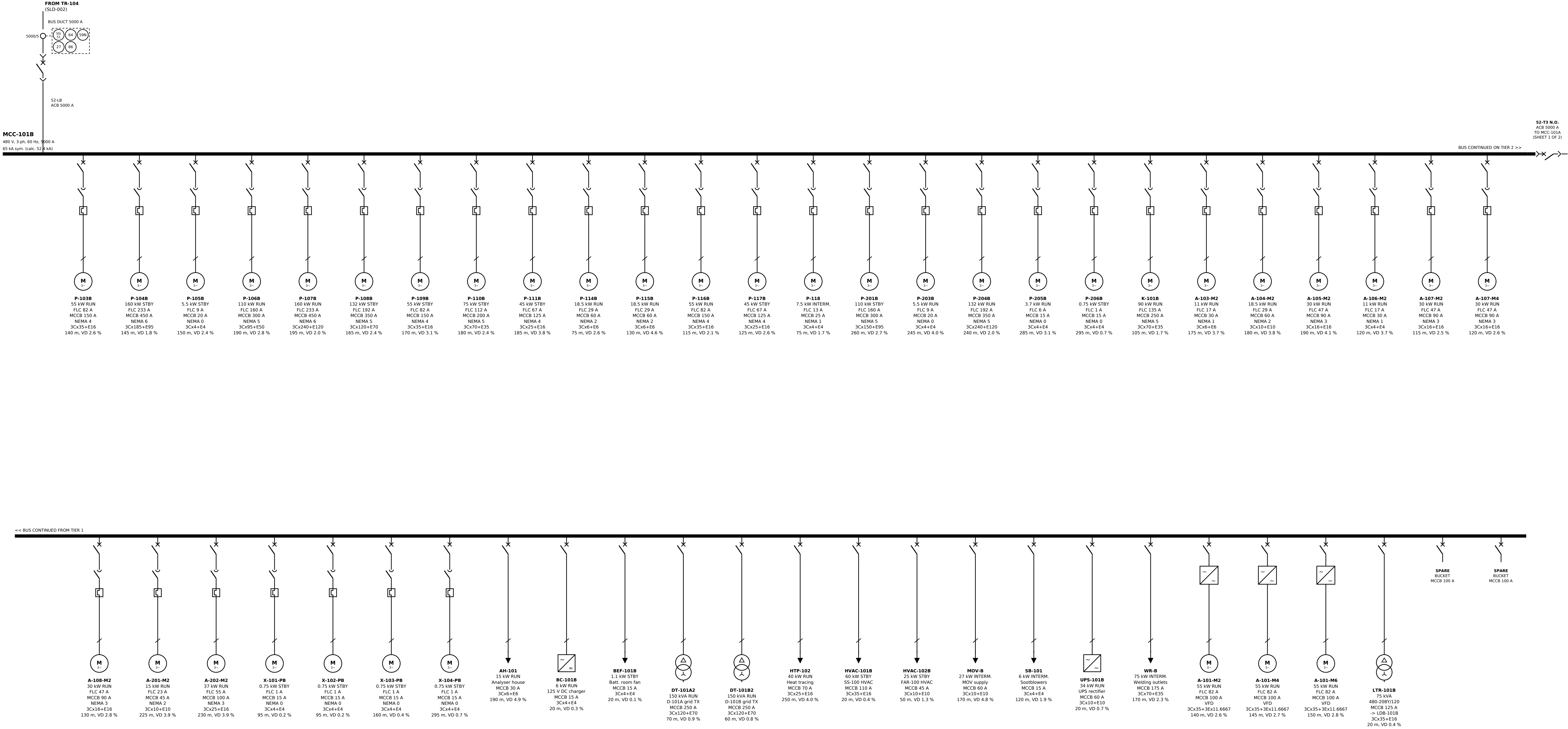


MCC-101A SUMMARY	VALUE
Feeders / buckets	50 (35 motor, 2 spare)
Connected load	1,769 kW
Continuous / Interm. / Standby	1,084 / 127 / 559 kW
Maximum demand (C+0.3I+0.1S)	1,178 kW / 1,342 kVA, PF 0.88
Both sections on one TR (tie closed)	2,676 kVA vs TR 3000/3750 kVA ONAN/ONAF
Largest DOL motor start - bus dip	3.5 % (limit 10 %)

- NOTES**
- MCC-101A: 480 V, 3-ph 3-W, 5000 A Cu bus, 65 kA sym.; ACB incomer/tie in LV switchgear section (IEEE C37.20.1), MCC sections UL 845, withdrawable buckets.
 - Motor buckets: MCCB (thermal-magnetic, <= 250 % FLC per NEC 430.52) + contactor + electronic OL, IEC 60947-4-1 Type 2 coordination; 120 V control transformer; DCS hardwired start/stop/run/trip.
 - Motors >= 30 kW; 4-20 mA motor current to DCS. VFDs: 6-pulse + passive filter (THDI <= 8 %).
 - Cable: Cu/XLPE/SWA/PVC 0.6/1 kV (IEC mm2). Sizes per CFU-000-EL-CBL-001 (ampacity x 1.25 FLC, VD <= 5 % running, <= 15 % at motor terminals during start, SC withstand via MCCB let-through).
 - Pump A/B pairs on opposite MCC sections; RUN/STBY = normally running unit for demand calculation.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
SINGLE LINE DIAGRAM			
480 V SWITCHGEAR / MCC-101A			
DISCIPLINE		STAGE	SCALE
ELECTRICAL		FEED	NTS
			1 OF 2
DRAWN		CHECKED	APPROVED
CLAUDE CODE		-	-
			DATE
			2026-10-02
DRAWING NUMBER			REV
CFU-000-EL-SLD-003			A



MCC-101B SUMMARY	VALUE
Feeders / buckets	51 (36 motor, 2 spare)
Connected load	1.779 kW
Continuous / Interm. / Standby	1.080 / 124 / 575 kW
Maximum demand (C+0.3I+0.1S)	1.175 kW / 1.335 kVA, PF 0.88
Both sections on one TR (tie closed)	2.676 kVA vs TR 3000/3750 kVA ONAN/ONAF
Largest DOL motor start - bus dip	3.5 % (limit 10 %)

NOTES

1. MCC-101B: 480 V, 3-ph 3-W, 5000 A Cu bus, 65 kA sym.; ACB incomer/tie in LV switchgear section (IEEE C37.20.1), MCC sections UL 845, withdrawable buckets.

2. Motor buckets: MCCB (thermal-magnetic, <= 250 % FLC per NEC 430.52) + contactor + electronic OL, IEC 60947-4-1 Type 2 coordination; 120 V control transformer; DCS hardwired start/stop/run/trip.

3. Motors >= 30 kW; 4-20 mA motor current to DCS. VFDs: 6-pulse + passive filter (THDI <= 8 %).

4. Cable: Cu/XLPE/SWA/PVC 0.6/1 kV (IEC mm2). Sizes per CFU-000-EL-CBL-001 (ampacity x 1.25 FLC, VD <= 5 % running, <= 15 % at motor terminals during start, 5C withstand via MCCB let-through).

5. Pump A/B pairs on opposite MCC sections; RUN/STBY = normally running unit for demand calculation.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
SINGLE LINE DIAGRAM 480 V SWITCHGEAR / MCC-101B			
DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	NTS	2 OF 2
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-EL-SLD-003			A